



USN

NMAM INSTITUTE OF TECHNOLOGY, NITTE

(An Autonomous Institution affiliated to VTU, Belagavi)

Seventh Semester B.E. (E&C) (Credit System) Degree Examinations

Supplementary Examinations - December 2025

21ECE133 / 20ECE73 / 16ECE74 - CRYPTOGRAPHY

Duration: 3 Hours

Max. Marks: 100

Note: 1) Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.

2) Missing data are to be assumed.

Unit – I		Marks	BT*	CO*	PO*
1.	a) Define Cryptography. Briefly explain the Active and Passive attacks.	10	L*2	1	1
	b) Explain the Euclidean algorithm to find the GCD of two integers.	5	L2	2	1
	c) Show 8 and 15 are relatively prime because the positive divisor of 8 are 1,2,4 and 8, and the positive divisor of 15 are 1,3,5 and 15.	5	L3	2	1
2.	a) Briefly explain the Model for Network Security.	8	L2	1	1
	b) Solve the following using the Chinese Remainder Theorem. i) $X \equiv 2 \pmod{3}$ ii) $X \equiv 3 \pmod{5}$ iii) $X \equiv 2 \pmod{7}$	8	L3	2	1
	c) Explain the difference between cryptography and steganography. Give an example where steganography would be preferred over cryptography.	4	L2	1	1
3.	a) Explain the different transposition techniques in security.	10	L2	1	1
	b) Encrypt the message "DEFENDTHEBASE" using the key "ARMY" with the Vigenère cipher.	6	L3	1	1
	c) Solve $\log_2 9 \pmod{11}$ using discrete logarithm.	4	L3	2	1
Unit – II					
4.	a) Describe the overall scheme for DES algorithm and its silent feature.	10	L2	3	1
	b) Using knapsack algorithm for a super increasing sequence $w=\{2,7,11,21,42,89,180,354\}$ $q=881$ $r=588$. Show how message 01110011 is encrypted. Also show the decryption process of the cipher message.	10	L3	4	1
5.	a) Explain the RSA algorithm in detail. Given two distinct prime numbers $p=13$ and $q=19$, perform the following: i) Calculate the modulus n used in RSA. ii) Compute $\phi(n)$ the Euler's totient function for n . iii) Choose an encryption exponent $e=7$ and verify that it is coprime with $\phi(n)$.	12	L3	4	1
	b) In the ElGamal cryptosystem, given the prime number $p=23$ and the generator $g=5$, choose a private key $x=6$. Calculate the public key y where $y=g^x \pmod{p}$. Show your calculations for finding y .	8	L3	4	1
6.	a) Explain the Diffie-Hellman key exchange algorithm. Show that the keys generated at sender side and receiver side are same.	10	L2	4	1
	b) Explain the modes of operation of block ciphers.	10	L2	3	1
Unit – III					
7.	a) Explain message authentication requirements, how MACs are used, and compare SHA-1 and MD5 in terms of security and application.	10	L2	5	1
	b) Explain Digital Signature algorithm in detail.	10	L2	5	1
8.	a) What is quantum cryptography, and how does it differ from classical cryptography? Briefly discuss on Quantum computing.	10	L2	5	1
	b) Explain the fundamental components and security features of blockchain technology.	10	L2	5	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
